Brownian motion and stochastic integrals, viewed from Ito's isometry and Girsanov's theorem [DRAFT. Thoughts only]

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There are two dual approaches to this that I've observed. One concerns Stochastic processes, the other — distributions. In this article I try to carry out a pedagogical presentation of material studied at uni, i.e. in a way that feels as "natural" as possible rather than forced.

Contents

1 Ultra Basics

1.1 Brownian motion

Brownian motion is defined either

1. as a stochastic process $(W_t)_{t\geq 0}$ satisfying three properties:

(a) $W_0 = 0$

(b) W_t is normally distributed with variance t; and all increments (of the same length) $\Delta W_{s,t}$ share that same distribution;
Later I will write ΔW_t to mean $\Delta W_{0,t}$, so just W_t, but written in the increment style to stress that we are interested in increments of a particular length. By stationarity, expected values then are the same for increments not started at zero, too.

(c) For $s < t$, $W_s \perp \Delta W_{s,t}$, i.e. independent increments

2. via a limiting distribution on the space of continuous functions $C([0,1], \mathbb{R}^n)$ of the distributions of random walks on uniform grid $\{t_i := i/n\}_{i=0}^n$ with jump $1/\sqrt{n}$:

   For each $n > 0$, let $(\xi_i)_{i=0}^n$ be a standard random walk where the jump $\Delta\xi_i = \xi_{i+1} - \xi_i$ is either $1/\sqrt{n}$ or $-1/\sqrt{n}$ with equal probability 0.5. We think of time here as flowing discretely in $\{t_i := i/n\}_{i=0}^n$.

   Now conceptually[1], the distribution of the random vector $(\xi_1 \ldots \xi_n)$ converges to a distribution on the space of continuous functions $C([0,1], \mathbb{R})$.

   It is trivial[2] now to define Brownian motion on the probability space $\Omega := C([0,1], \mathbb{R})$ with that limit distribution as a measure by just projecting each sample (trajectory, function $[0,1] \to \Omega$)

$$\text{for } \omega \in \Omega : \quad W_t(\omega) := \omega_t.$$

The former is the definition useful for applications/proving properties of the brownian motion, while the latter gives an idea how it came to be. As long as we believe in those two definitions and trust that they're equivalent, we're good to proceed.

There are two free-hanging postulates in these definitions that are important to understand first, though:

1. If we believe that W_t is a process of stationary and independent increments, what is left to motivate is

   why should the variance of its increments ΔW_t be chosen to be t?
   (or the standard variation — $\sqrt{t}$)

   (one could also ask why the increments' distribution should be Gaussian, buuut no time to ponder on that now).

2. On the other hand, if we believe that W_t is a limit of distributions of random walks, what is unclear is

   why should the jumps at epoch (n), be $1\sqrt{n}$?

[1] In N.V. Krylov's book "Introduction to the theory of random processes" this is stated and proven rigorously (ξ is first extended from the discrete points $\{t_i\}_i$ continuously to the whole interval $[0,1]$).

[2] Krylov's book here also proves that this definition makes sense. That is not as trivial.

1.2 Why jump with $\pm\frac{1}{\sqrt{n}}$ and why $\operatorname{Var}\Delta W_t = t$

Why variance is t (and not any other function of t) is explained by the $\pm\frac{1}{\sqrt{n}}$ jump in the approaching distributions, but without motivating that jump, the explanation is free-hanging.

Let $\eta_k^{(n)}$ be independent identically distributed with values -1 and 1 with probability 0.5 each. Set a random walk with jump $\alpha_n > 0$:

$$\xi_0^{(n)} := 0, \quad \xi_{k+1}^{(n)} := \xi_k^{(n)} + \alpha_n \eta_k^{(n)}$$

Put a grid $\{t_0^{(n)}, \ldots, t_n^{(n)}\}$ with $t_k = k/n$. Brownian motion will be the "limit random walk" as n tends to infinity and the grid refines to the full interval $[0,1]$.

For a continuous variable $t \in [0,1]$, the nearest grid point to the left of t at epoch n is $t^{(n)} := \lfloor nt \rfloor / n$, so the k at which $\xi_k^{(n)}$ can be evaluated is $\lfloor nt \rfloor$. Hence put

$$\xi_t^{(n)} := \xi_{\lfloor nt \rfloor}^{(n)}$$

Now

$$\operatorname{Var}\xi_k^{(n)} = \operatorname{Var}\sum_{i=1}^{k} \alpha_n \eta_i^n \overset{\perp}{=} \sum_{i=1}^{k} \operatorname{Var}\alpha_n \eta_k^n = \alpha_n^2 k$$

so

$$\begin{aligned}
\operatorname{Var}\xi_t^{(n)} &= \operatorname{Var}\xi_{\lfloor nt \rfloor}^{(n)} = \alpha_n^2 \lfloor nt \rfloor \\
&= t^{(n)} \cdot n\alpha_n^2 \\
&= t^{(n)} \cdot \left(\alpha_n \sqrt{n}\right)^2
\end{aligned}$$

where $t^{(n)}$ is the closest grid point to the left of t as before at epoch n, and is clearly stable around a value in $(0,1)$ for large n. The problematic part is the other multiple, $\left(\alpha_n \sqrt{n}\right)^2$:

If we want that for high n...	...then α_n has to tend to zero $\alpha_n \searrow 0$
$\operatorname{Var}\xi_n^{(n)}$ stays bounded	at least as fast than $1/\sqrt{n}$
$\operatorname{Var}\xi_n^{(n)}$ does not vanish[3]	no faster than $1/\sqrt{n}$

Hence to construct a brownian motion with finite but non-vanishing variation, the random walks it is built from must use a jump $\eta_k^{(n)} = \pm\alpha_n$ of order $1/\sqrt{n}$. We simply pick[4] $\alpha_n = 1/\sqrt{n}$, which gives

$$\operatorname{Var}\xi_t^{(n)} = t^{(n)} \cdot \cancel{\left(\alpha_n \sqrt{n}\right)^2} \xrightarrow[n\to\infty]{} t.$$

This is how we choose $1/\sqrt{n}$ for the random walk jumps $\eta^{(n)}$, and variance t (deviation $\sqrt{t}$) for the Gaussian increments ΔW_t.

[3] so that the limiting distribution is not a point mass at 0

[4] We could pick any constant and would get the limit of the variation scaled accordingly; a madman might choose $\alpha_n = \frac{\sin n}{\sqrt{n}}$ but then their $\operatorname{Var}\xi_t^{(n)}$ would not converge to a single value.

1.3 Stochastic integral for simple functions

In a Lebesgue-Stieltjes integral style we can integrate simple predictable functions against the brownian motion.

Remember that the Lebesgue-Stieltjes integral against a function (of bounded variation) g,

$$\int_X f \mathrm{d}g$$

works similarly to the standard Lebesgue integral but instead of using a *measure* for the floor increments ($\mu(x-y)$) when infinitesimally multiplying the integrated function, it uses the provided function g's increment, $\Delta g = g(x) - g(y)$, to "measure" it.

A simple process is a process taking finitely many values that we denote $f_1, \ldots, f_k$, i.e. it is expressible as a finite sum

$$f^\omega(t) = \sum_n f_n^\omega \mathbf{1}_{[t_n, t_{n+1}]}(t)$$

So we define the stochastic integral against the brownian motion $\mathrm{d}W_s$ of simple predictable functions as

$$\int_0^t f(s) \mathrm{d}W_s := \sum_n f_n \Delta W_n$$

2 Square of Δoutput is expected to be the Δinput

There are a range of facts that I summarize under this section's title. The chaos of the material is vastly reduced by viewing many results as an instance of this pattern.

2.1 Of W_t itself

2.1.1 Discretely

The second view of Brownian motion above also tells us that considering one jump of the random walk, say

$$\text{time: } t_i \to t_{i+1}$$
$$\text{space: } \xi_i \to \xi_{i+1}.$$

With the choice $\Delta t_i = 1/n$, $\Delta \xi_i$ is either $\frac{1}{\sqrt{n}}$ or $-\frac{1}{\sqrt{n}}$. Thus the square $\Delta \xi_i^2$ is most definitely $\frac{1}{n}$. That is,

$$\Delta \xi_i^2 = \Delta t_t$$

— the square of the output's increment equals the input increment.

2.1.2 Continuously

In the limiting distribution $\Delta W_{s,t}$ is definitely does not square to just $(t-s)$, but it is interesting that its mean *does*:

$$\mathbb{E}\left(\Delta W_t^2\right) = t.$$

We see this from the property charaterising the distribution of the (non-squared) increment ΔW_t:

$$\begin{aligned}\mathbb{E}\left(\Delta W_t^2\right) &\equiv \int_\Omega \Delta W_t^2 \mathrm{d}\mathbb{P} = \int_\mathbb{R} y^2 \mathrm{d}\mathbb{P}_Z \\ &= \int_\mathbb{R} y^2 \Phi(y)\mathrm{d}y \overset{!}{=} t \int_\mathbb{R} \Phi(y)\mathrm{d}y = t\end{aligned}$$

where in $\overset{!}{=}$ we used $\Phi(y) =$(σ^2 denoting the variance of the normal variable, i.e. here just t; also omitting integral signs)

$$\begin{aligned}y^2 e^{-y^2/2\sigma^2}\mathrm{d}y &= -2\sigma^2 y e^{-y^2/2\sigma^2}\mathrm{d}y^2 / -2\sigma^2 \\ &= -y\mathrm{d}e^{-y^2/2\sigma^2} \\ &= -ye^{-y^2/2\sigma^2} + e^{-y^2/2\sigma^2}\mathrm{d}y.\end{aligned}$$

2.2 Of the Itô integral for simple functions

Above we computed the expected square of the raw increments ΔW_t. In a martingale-transformation fashion we can consider the "transformed" increments $f\Delta W$ and their integral's expected square:

$$\begin{aligned}\mathbb{E}\left[\left(\int_0^t f(s)\mathrm{d}W_s\right)^2\right] &\equiv \mathbb{E}\left[\left(\sum\nolimits_n f_n \Delta W_n\right)^2\right] \\ &\overset{\text{lin}}{=} \mathbb{E}\left[\sum_n f_n^2 \Delta W_n^2\right] + \mathbb{E}\left[\sum_{n\neq m} f_n f_m \Delta W_n \Delta W_m\right] \\ &\overset{\perp}{=} \sum_n \mathbb{E}\left[f_n^2\right] \underbrace{\mathbb{E}\left[\Delta W_n^2\right]}_{\Delta t} + \cancel{2\sum_{n<m} \mathbb{E}\left[f_n f_m \Delta W_n\right] \underbrace{\mathbb{E}\left[\Delta W_m\right]}_{0}} \\ &= \mathbb{E}\left[\sum_n f_n^2 \Delta t\right] \\ &\equiv \mathop{\mathbb{E}}_{\omega}\left[\int_0^t f_\omega^2(s)\mathrm{d}s\right]\end{aligned}$$

which I roughly interpret as

> The expected **square** of the modified (by f_t) output (of ΔW_t) equals the **linear** integral of the output's modification's square.

In a way, the ΔW_t^2 collapses inside the integral to Δt, while the f^2 multiple remains as a squre.

This peculiar fact gives the Ito integral the name "Ito's Isometry" (for the very special case of simple functions).

2.3 Itô of a deterministic process

It is a bit of a stretch, but for a deterministic processes the stochastic integral can also be seen to exhibit this output-is-square-of-input pattern.

If θ_s is deterministic, i.e. θ is just a function $\theta : [0,T] \to \mathbb{R}$, then $\int \theta_s \mathrm{d}W_s$ is normal with variance $\int \theta_s^2 \mathrm{d}s$. That is near-obvious for *simple* θ, i.e. when $\theta(t) = \theta_i$ over intervals $t \in [t_i, t_{i+1}]$, because then the Itô integral

$$\int_0^t \theta_s \mathrm{d}W_s = \sum_i \theta_i \Delta W_i$$

is a weighted sum of the independent normal variables ΔW_i, and that is again normal. The variance is

$$\begin{aligned}
\operatorname{Var} \int_0^t \theta_s \mathrm{d}W_s &= \sum_i \operatorname{Var} \theta_i \Delta W_i \\
&\overset{\perp}{=} \sum_i \theta_i^2 \operatorname{Var} \Delta W_i \overset{W}{=} \sum_i \theta_i^2 \Delta t_i \\
&= \int_0^t \theta_s^{\mathbf{2}} \mathrm{d}s
\end{aligned}$$

the "square-weighted" input. For $\theta = 1$, this recovers $\operatorname{Var} W_t = t$, while for other non-uniform weights, that t is re-weighted with the square of the integrand to $\int_0^t \theta_s^2 \mathrm{d}s$.

3 Itô's formula

Ito's formula is just an application of Tailor's formula for a stochastic version of the Fundamental theorem of calculus.

Tailor's formula

$$\Delta f(x) \equiv f(x + \Delta x) - f(x) = f'(x)\Delta x + \frac{1}{2} f''(x)\Delta x^2 + \text{rest}$$

with $x = W_t$ and the principle $(\Delta W)^2 \sim \Delta t$ applied to the second order term reads

$$\begin{aligned}
\Delta f(W_t) &= f'(W_t)\Delta W + \frac{1}{2} f''(W_t)\Delta W^2 + \text{rest} \\
&\sim \frac{1}{2} f''(W_t)\Delta t + f'(W_t)\Delta W + \text{rest}
\end{aligned}$$

Summing that at the points of a partition $t_0, \ldots, t_n$ and taking a limit this turns out to be,

$$df(W) = \frac{1}{2} f''(W)dt + f'(W)dW.$$

One day I want to write this down rigorously, but no time today for it.